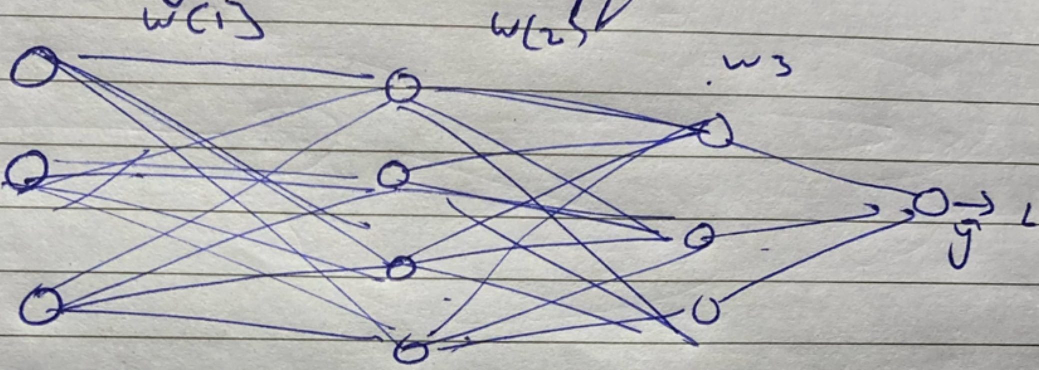
Vanishing Gradient descent

Jb hom gradien descent nikalte eg. hidden layes mein Activation layer hamne sigmoid liy or kaffi complex Neural Network liya, sigmodi (0-1) ~~unit~~ whiawj's hame kaff kam update mi'ly



$$\frac{\partial L}{\partial w'_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{d\hat{y}}{\partial z} \times \frac{d\hat{y}}{dO_{31}} \times \frac{dO_{31}}{\partial_{21}} \times \frac{\partial O_{21}}{dw'_{11}}$$

because of small multiplication & afterwards multiplying with learning rate it make very small &

$w_n = w_{old} - \eta \frac{dL}{dw}$

it make neglible change

1 → So we use ReLU Activation layer $\max(0, z)$ more the range

2 → Decrease Complexity of Neural Network, making less hidden layers

if you data doesn't have complexity then only use it, agar layers kam toh complexity kaise handle karenge

3) proper weight init

[4] Batch Normalization layer

[5] using Residual Network